Build AI Agents with the OpenAI Agents SDK

Mode 2 Phase 2 (Build Workers), Chapter 1/3 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/build-agents-crash-course

EN: The rule for this whole course: every agent bug is either a STATE bug or a TRUST bug. Agent = LLM + instructions + tools; Runner runs the loop (model -> tools -> results -> model...).
Roman Urdu: Poore course ka rule: har agent bug ya STATE bug hai, ya TRUST bug. Agent = LLM + instructions + tools; Runner loop chalata (model -> tools -> results -> model...).

1. Three Primitives + the Agent Loop

- Agent, Runner (run_sync/run/run_streamed), @function_tool (type hints+docstring -> JSON schema). Default model gpt-5.4-mini.
- max_turns caps the loop (MaxTurnsExceeded). You're not IN the loop -- you control it upstream (instructions/tools/guardrails) and downstream (result).

2. Sessions, Streaming, Tools, Handoffs

- Runner.run_sync is STATELESS -- turn 2 doesn't remember turn 1 unless you add a Session (SQLiteSession). Cost consequence: every turn re-bills the full history.
- Function tool type hints ARE the model's documentation (Literal[...] forces exact values). Handoffs transfer control to a specialist agent -- cost: each handoff adds >=1 extra model call.

3. Safety, Observability, Routing, Approval

- Guardrails: input/output/tool, parallel (default) vs blocking. Tracing: trace from Day 1 -- microsecond overhead vs hours of blind debugging.
- Model routing: economy vs frontier tiers, cost gap 10x+. Human approval (needs_approval=True): approval is NOT a substitute for sandboxing, and vice versa -- you need both.

4. Sandboxing + Persistence

- The model is NOT a reliable safety boundary by itself (it usually refuses, not always). Trap: only BUILT-IN capabilities get sandboxed -- your own @function_tool bodies still run in your Python process.
- R2 mount gives durable storage; the sandbox container itself dies fast on idle timeout -- only /workspace/data (mounted) survives.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Agent = LLM + instructions + tools	The static definition the Runner executes.	Static definition jo Runner execute karta.
Runner	Runs the agent loop: model -> tools -> results -> model.	Agent loop chalata: model -> tools -> results -> model.
Session (stateless fix)	Adds memory across turns (SQLiteSession) -- without it, each run forgets prior turns.	Turns ke across memory add karta (SQLiteSession) -- bina iske har run pichle turns bhool jata.
Guardrail	Input/output/tool checks, parallel or blocking.	Input/output/tool checks, parallel ya blocking.
Approval != Sandboxing	Human approval and execution sandboxing solve different risks -- you need both.	Human approval aur execution sandboxing alag risks solve karte -- dono chahiye.

Quick Self-Check

Question	Answer
What's the rule-in-one-line for debugging this course's agents?	Every agent bug is either a state bug or a trust bug.
Why is Runner.run_sync alone a 'classic state bug' waiting to happen?	It's stateless -- turn 2 won't remember turn 1 unless you explicitly add a Session.
Why isn't the model itself a reliable safety boundary?	It usually refuses unsafe requests, but not always -- and only built-in capabilities get sandboxed, not your own function-tool code.
Why must tracing start on Day 1, unlike other safety primitives?	It has near-zero overhead and saves hours of blind debugging later -- unlike other safety add-ons, there's no reason to delay it.

Where This Leads / Sources

Leads to: Building a Digital FTE, Give Your AI Agent a Nervous System. Full worked example (2-stage: local + sandboxed) in Part 4; cost discipline in Part 5.

Source: docs/build-agents-crash-course/ (this repo), content originally via Zia Tutor AI connector.